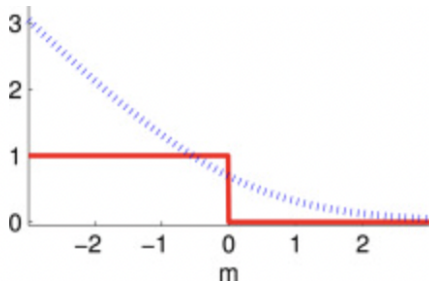


MultiClass Classification

Procheta Sen



Classification algorithms

Multiclass classifiers

- ▶ k-NN
- ▶ Naive Bayes

Binary classifiers

- ▶ Perceptron
- ▶ Logistic regression

How to turn a binary classifier to a multiclass classifier?

Given binary classification algorithm $\mathcal{A}$ we want to design a meta-algorithm that uses $\mathcal{A}$ to make k -class predictions.

Two strategies

- ▶ **One-vs.-one** approach
- ▶ **One-vs.-rest** approach

One-vs.-one approach

1. For each pair of classes i and j , use training objects from classes i and j to train algorithm $\mathcal{A}$ to distinguish between objects in classes i and j . Denote the obtained classifier $A_{i,j}$.

2. This results in

$$\frac{k(k-1)}{2}$$

prediction models.

3. Applying the prediction model $A_{i,j}$ to an incoming object $\bar{X}$ is interpreted as voting. $A_{i,j}$ votes either +1 for $\bar{X}$ to be in class i , or $A_{i,j}$ votes +1 for $\bar{X}$ to be in class j .
4. For an incoming object $\bar{X}$, apply all prediction models one by one.
5. The class label with the most votes is declared as the winner.

Frame Title

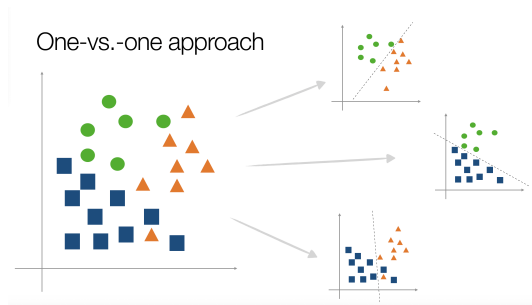


Figure: Caption

One-vs.-one approach: drawbacks

- ▶ There might be ambiguity if some classes get the same number of votes (if the binary classifier $\mathcal{A}$ can produce a confidence score, it can be used to break ties).

One-vs.-rest approach

In this approach we assume that the binary classification algorithm $\mathcal{A}$ can output a numeric score representing its “confidence” that an object belongs to a particular class.

1. For each class i , train the binary classifier $\mathcal{A}$ with the objects of class i treated as positive samples and all other objects as negative samples. Denote the obtained classifier A_i .
2. This results in k prediction models.
3. For an incoming object $\bar{X}$, apply all prediction models $A_1, A_2, \dots, A_k$.
4. Output for object $\bar{X}$ the class label y corresponding to the model with the highest score, i.e.,

$$y = \arg \max_{i \in \{1, 2, \dots, k\}} A_i(\bar{X})$$

One Vs. Rest Approach

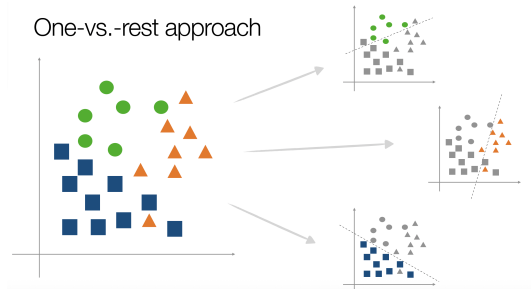


Figure: Caption

One-vs.-rest approach

The choice of the numeric score depends on the classifier at hand.

1. For Perceptron: the activation score

$$a = b + \overline{W}^T \overline{X}$$

2. For Logistic regression: $\sigma(a)$, where

$$a = b + \overline{W}^T \overline{X}$$

One-vs.-rest approach: drawbacks

- ▶ the scale of the confidence scores may differ between the binary classifiers
- ▶ the binary classifiers are trained on unbalanced datasets: usually the set of negative objects will be much larger than the set of positive objects